MARK IVAN UGALINO

www (dot) markivanugalino (dot) com ◇ ugalino (at) umd (dot) edu

EDUCATION

University of Maryland, College Park — Fall 2023 — *present*
Doctor of Philosophy in Astronomy, *PhD candidate since Fall 2025*
Department of Astronomy

University of Massachusetts, Dartmouth — 2021 — 2023
Master of Science in Physics
College of Engineering

University of the Philippines, Diliman — 2018 — 2020
Master of Science in Physics
National Institute of Physics

University of the Philippines, Diliman — 2013 — 2018
Bachelor of Science in Physics
National Institute of Physics

Quezon City Science High School — 2009 — 2013
High school diploma

SKILLS

Computer Languages	Python, Fortran, C, C++ *knowledgeable* in Julia and R
Software & Tools	ART (C, Kravstov et al), RAMSES (Fortran, Teyssier), Arepo (C, Springel), SWIFT (C, Schaller et al), FLASH (Fortran, Fryxell et al), MESA (Fortran, Paxton et al), yt (Python, Turk et al), SuperNu (Fortran, van Rossum, Wollaeger), Torch (Fortran, Timmes)
HPC Systems	TACC Stampede2, UMassD CARNIE, UMD Zaratan

MENTORING

Nar Jasen Dela Cruz (BS Physics '26, University of the Philippines Baguio, co-adviser *now* Academia Sinica Institute of Astronomy and Astrophysics summer student)
Project: *Investigating lunar formation scenarios using SPH*

Jin Young Kim (Wake Technical Community College *through* GRAD-MAP Summer Scholars 2025, *now* BS Physics, University of North Carolina Chapel Hill)
Project: *Magnetic fields in galaxies*

Emau Argueta (BS Mathematics, University of Maryland)

Prabodha Mudalige (MS Physics, UMass Dartmouth *now* PhD Physics, UT Knoxville)
Project: *Viscous evolution of astrophysical disks*

Vrutant Mehta (MS Physics, UMass Dartmouth *now* PhD Engineering and Applied Science, UMass Dartmouth)

Jack Sullivan (BS Physics, UMass Dartmouth *now* PhD Physics, University of Connecticut)

GRANTS AND RECOGNITIONS

Do Good Campus Fund Grant (USD 6,000.00) — 2026
"GRAD-MAP: A unique model for graduate student-led mentoring and outreach in astronomy and physics", (with Yash Gursahani, lead)

Jacob K. Goldhaber Travel Grant (USD 600.00) Spring 2026
Awarded by UMD Graduate School for participation in "RAMSES User Meeting (RUM) 2026"

Maryland Space Grant Consortium Activity Grant (USD 28,340.00) 2026
"GRAD-MAP: A unique model for graduate student-led mentoring and outreach in astronomy and physics", (Co-PI, with Yash Gursahani and Benedikt Diemer)

Jacob K. Goldhaber Travel Grant (USD 250.00) Fall 2025
Awarded by UMD Graduate School for participation in "Magnetic Fields and Cosmic Rays across Diverse Scales: What's Next"

2023 NASA FINESST Research Grant (USD 148,826.00) *declined*
Among first 27 selected for award out of 262 submitted proposals (10.3% award rate)

Graduate Dean's Fellowship (USD 10,000.00) Fall 2023
Department of Astronomy, University of Maryland College Park

Chancellor's Centennial Engineering Scholarship Spring 2023
University of Massachusetts Dartmouth

Graduate Research Award 04/27/2022
Department of Physics, University of Massachusetts Dartmouth

FIP Distinguished Student Award 02/2022
American Physical Society

Gawad Direktor para sa Natatanging Bagong Guro 12/07/2018
National Institute of Physics, UP Diliman

- The award was given in recognition of the exemplary performance of a newly hired junior faculty of the institute.

Gawad Direktor para sa Natatanging Discussion Teacher 12/07/2018
National Institute of Physics, UP Diliman

- This award is given in recognition of the exemplary performance of a junior faculty member as a discussion teacher for lecture classes offered by the institute.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant 01/2024–present
Department of Astronomy, University of Maryland College Park

- Leads a computational project that explores how magnetic fields evolve in galaxies through large-scale simulations using codes like ART, Arepo, and RAMSES

Graduate Teaching Assistant 08–12/2023
Department of Astronomy, University of Maryland College Park

- Teaching assistant for Prof. Lee Mundy's ASTR 101 (Fall 2023)
- Led weekly discussions with a non-technical audience on foundations of astronomy and astrophysics
- Supervised and graded laboratory work of a class with 20 students at a time

Graduate Student Intern 06/2023–08/2023
Los Alamos National Laboratory

- Participant during the 2023 installment of the LANL Co-Design Summer School (est. 2011, http://lanl.github.io/cdss/) where we developed a performance portable code for thermonuclear supernovae

Graduate Teaching Fellow 09/2022–05/2023
Department of Physics, UMass Dartmouth

- Designed (as Instructor on Record) an introduction to astronomy course for a non-technical audience, with topics ranging from history, stellar evolution, galaxies, cosmology and indigenous astronomy. Graded exams and class output designed to synthesize lectures delivered in class.

Graduate Research Assistant 05/2021–08/2022
Department of Physics, UMass Dartmouth

- Led computational projects on type Ia supernovae that explore (1) a first-principles turbulently-driven explosion mechanism in near-Chandrasekhar mass white dwarfs, (2) the magnetized evolution of white dwarf merger remnants.
- Co-led a paper that illustrated how the the isotope Ti-44 can be used to constrain the progenitor of a type Ia supernova event (*see* Publications)

Graduate Teaching Assistant 01/2021–05/2021
Department of Physics, UMass Dartmouth

- Taught online recitation and laboratory courses in Classical Mechanics

Instructor 1 08/2018–12/2020
National Institute of Physics, UP Diliman

- Taught in-person and online lectures and laboratory classes in Classical Mechanics, Electromagnetism, Physical Electronics, and Computational Physics
- Designed exams and in-class exercises for introduction to classical mechanics and electromagnetism classes
- Co-organized events for the department, served as course group leader for electromagnetism labs

SERVICE

UMD GRAD-MAP Co-Lead 01/2024–08/2026
Department of Astronomy, University of Maryland College Park

- Co-authored a successful Do Good Institute Grant proposal (USD 6,000.00) with Yash Gursahani, where Mr Gursahani is the lead author
- Co-authored a successful Maryland Space Grant Consortium proposal (USD 28,340.00) with Yash Gursahani and Benedikt Diemer
- Prepared yearly budgets, materials for programs, and other logistics
- Organized Journal Clubs led by our summer scholars, provided feedback and moderated each session
- Developed our website at `https://www.umdgradmap.org/` using Jekyll
- Moderates and creates content for our social media accounts on Instagram and LinkedIn
- Moderates professional skills workshops on how to give elevator pitches, coding in Python, creating documents with LaTeX and how to prepare and give a scientific talk
- Served as a research mentor to one student (Jin Young Kim, *now at* UNC Chapel Hill)

Graduate Student Admissions Volunteer 02/2024 - present
Department of Astronomy, University of Maryland College Park

- Co-leads the prospective student visit planning committee for 2027 with Sander Somers
- Participated in prospective student interviews from 2024 to 2026

Graduate Student Council Representative 03/2026 - present
Department of Astronomy, University of Maryland College Park

- Represents 3rd year graduate students in our department student body

Reviewer (theoretical physics) 2019–2023
Proceedings of the Samahang Pisika ng Pilipinas Physics Conference

University of the Philippines Astronomical Society 2015–present
Associate Member, Education and Research Coordinator (2017)

PUBLICATIONS

First-Principles Turbulence-Driven Deflagration-to-Detonation Transition Mechanism for Near-Chandrasekhar Mass White Dwarf Progenitors
K Patel, R Fisher et al, **incl MI Ugalino** (2026) *ApJ* (in press)

Using ^{44}Ti Emission to Differentiate Between Thermonuclear Supernova Progenitors
D Kosakowski, **MI Ugalino**, R Fisher, et al (2023), *MNRAS Letters* 519(1), Pages L74–L78

Steady-state density perturbations induced by a point mass in a finite cylinder.
MI Ugalino, MFI Vega (2020) *Proceedings of the 38th Samahang Pisika ng Pilipinas Physics Congress*

Density perturbations in a collisional fluid induced by a particle on a slightly-eccentric orbit.
MI Ugalino, MFI Vega (2018) *Proceedings of the 36th Samahang Pisika ng Pilipinas Physics Congress*

TALKS AND POSTERS

Investigating the numerical dependencies of the galactic dynamo ...(*contributed talk*) 04/29/26
RAMSES User Meeting 2026, Yonsei University, Seoul, South Korea

Investigating the numerical dependencies of the galactic dynamo ... (*invited talk*) 09/12/2025
University of Massachusetts Dartmouth, North Dartmouth, MA, U.S.A.

Investigating the numerical dependencies of the galactic dynamo ... (*invited talk*) 09/11/2025
Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, U.S.A.

Investigating the numerical dependencies of the galactic dynamo ... (*poster*) 09/10/2025
Magnetic Fields and Cosmic Rays across Diverse Scales: What's Next?, Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, U.S.A.

The Turbulent Lives of Galactic Magnetic Fields (*invited talk*) 08/11/2025
1st Philippine Space Science and Astronomy Research Conference, Manila, Philippines (online)

Physics Speaks: It's turtles all the way down: lessons we can derive from cosmology (*invited talk*) 2023
Ateneo LeaPs

Ares – Simulating Type Ia Supernovae on Heterogeneous HPC Architectures (*poster*) 11/12-17/2023
Supercomputing 2023, Denver, CO, U.S.A.

Turbulently-driven deflagration-to-detonation transition in near-Chandrasekhar mass white dwarfs (*invited talk*) 06/15/2023
Center for Theoretical Astrophysics, Los Alamos National Laboratory

Turbulently-driven deflagration-to-detonation transition in near-Chandrasekhar mass white dwarfs (*iPoster*) 06/07/2023
American Astronomical Society Meeting 242, Albuquerque, NM, U.S.A.

Three-dimensional simulations of turbulently-driven deflagration-to-detonation transition in near-Chandrasekhar type Ia supernovae (*contributed talk*) 08/15-19/2022
EuroWD22, Eberhard Karls Universität Tübingen, Tübingen, Germany

Turbulently-driven deflagration-to-detonation transition in near-Chandrasekhar mass white dwarfs (*contributed talk*) 04/12/2022
American Physical Society, New York City, NY, U.S.A.

Turbulently-driven deflagration to detonation transition in near-Chandrasekhar mass white dwarfs (*invited talk*) 01/31/2022
Massachusetts Institute of Technology, MA, U.S.A.

Late-time dynamical friction in finite disks (*invited talk*) 02/04/2021
University of Massachusetts Dartmouth, MA, U.S.A.

Steady-state density perturbations induced by a point mass in a finite cylinder (*contributed talk*) 10/19/2020
38th Samahang Pisika ng Pilipinas Physics Conference, Philippines

SCHOOLS ATTENDED

Burgers Program Summer School on Turbulence 06/03—07/2024
University of Maryland Burgers Program for Fluid Dynamics

LANL Co-Design Summer School 05–08/2023
Los Alamos National Laboratory, NM, U.S.A.

MESA Summer School 08/08–12/2022
University of California Santa Barbara, CA, U.S.A

NSF/APS-DPP GPAP Summer school on plasma physics for astrophysicists 06/07–11/2021
Swarthmore College (online)

Deciphering Dark Matter: From Galaxies to the Universe 09/14–25/2020
Institut Teknologi Bandung, Bandung, West Java, Indonesia (on-line)

PRESS

"GRAD-MAP Students, Mentors 'Learn From Each Other' "(August 20, 2025)

PROFESSIONAL REFERENCES

Prof. Benedikt Diemer
Adviser (PhD) and Associate Professor of Astronomy
`diemer (at) umd (dot) edu`
Department of Astronomy, UMaryland College Park

Prof. Robert Fisher
Adviser (MSc) and Professor of Physics
`robert.fisher (at) umassd (dot) edu`
Department of Physics, UMass Dartmouth

Prof. Ian Vega
Adviser (BS and MS) and Professor of Physics
`ivega (at) nip (dot) upd (dot) edu (dot) ph`
National Institute of Physics, UP Diliman

Prof. Bo Dong
Associate Professor of Mathematics
`bdong (at) umassd (dot) edu`
Department of Mathematics, UMass Dartmouth

Updated: June 2026